P Rupak Chand
Email – rchandblue@gmail.com
Number – 860-544-6305
LinkedIn- https://www.linkedin.com/in/rupakchand-paramkusam-04a65b292/

Professional Summary:

MLOps Engineer and Data Scientist with over 4 years of expertise in deploying and managing machine learning models, automating ML workflows, and orchestrating cloud-based infrastructures. Proficient in building scalable CI/CD pipelines for machine learning using tools such as Kubeflow, MLflow, and Airflow, alongside containerization with Docker and Kubernetes. Solid foundation in Data Science, including advanced machine learning algorithms (Random Forests, SVM, KNN, Deep Learning, NLP) and model tuning. Strong background in DevOps practices, leveraging Jenkins, Terraform, and Helm for continuous integration and infrastructure automation. Highly skilled in data analysis and data visualization using SQL, Pandas, and BI tools like Power BI, Tableau, and Looker. A proven ability to deliver data-driven insights, streamline operations, and optimize performance in AWS, GCP, and Azure cloud environments.

Technical Skills:

MLOps: Model deployment (CI/CD for ML), Model Monitoring, DataOps, Kubeflow, MLflow, Airflow, Docker, Kubernetes, Feature Store Management.
Data Science: Machine Learning (Regression, Classification, Clustering), Random Forests, SVM, KNN, Naive Bayes, K-Means, Deep Learning (CNN, RNN),LLM, NLP, Model Tuning & Optimization, Python, R, Scikit-learn, TensorFlow, PyTorch
DevOps: Continuous Integration (CI), Continuous Deployment (CD), Jenkins, Git, AWS, Azure, GCP, Terraform, Ansible, Helm
Data Analysis: SQL, Pandas, Data Wrangling, Exploratory Data Analysis (EDA), Data Visualization (Matplotlib, Seaborn, Power BI, Tableau)
Programming Languages: Python, R, Bash, SQL, No-SQL
Cloud Platforms: AWS, GCP, Azure
Version Control: Git, GitHub, GitLab
Database Technologies: MySQL, MongoDB, PostgreSQL.
BI Tools: Tableau, Looker, GIT, Power BI, RapidMiner, SAS, Excel (Advance), MS Office Suite.

EDUCATION:

- Prist University | Bachelor of Science
- Sacred Heart University | MS in Business Analytics

PROJECTS

Customer Churn Prediction: Developed a machine learning model using scikit-learn to predict customer churn for a telecom company. Process included data cleaning, EDA, feature engineering, logistic regression modeling, and cross-validation. Achieved over 85% accuracy and visualized performance metrics with confusion matrix and ROC curve.

Professional Experience:

EXPEDIA,Remote,CT

Data Scientist/MLOps Engineer | Feb 2024 – Present

- Architected and implemented end-to-end MLOps pipelines, resulting in a 40% reduction in model deployment time and a 25% increase in model performance.
- Deployed machine learning models to production, ensuring they can scale to meet business needs.
- Implemented robust model monitoring and alerting systems, reducing model drift incidents by 30%.
- Implemented model monitoring and alerting systems, improving model reliability by 35% and reducing downtime by 50%
- Automated model retraining and deployment using CI/CD, Docker, Kubernetes, and cloud platforms (AWS, GCP, Azure).
- Managed LLMOps for Continuous Improvement of Generative Models: Established LLMOps best practices for managing the lifecycle of Generative AI models, including automated retraining based on performance metrics, dynamic fine-tuning, and version control to ensure the models stay relevant and effective over time.
- Design and maintain infrastructure for training and deploying models at scale.
- Managed cloud resources and GPU clusters for efficient training.
- Optimize infrastructure for cost and performance based on budget and speed requirements.
- Implement scalable ML pipelines with tools like Kubeflow and MLflow for tracking and monitoring models.
- Automate CI/CD pipelines for ML models using Jenkins, GitLab CI, and cloud services (AWS, Azure).
- Containerize ML applications with Docker and Kubernetes for consistent deployment environments.
- Worked with data scientists and engineers to deploy models using Airflow for workflow orchestration.
- Build data processing pipelines with Apache Spark and AWS Glue for training and inference.
- Created Python scripts to automate data ingestion and transformation processes, enhancing workflow efficiency and reducing manual errors.
- Used Terraform to automate cloud resource provisioning with infrastructure-as-code (IaC).
- Transition data science prototypes to production-ready models.
- Used tools like MLflow or DVC to track model versions, datasets, and results.

- Implemented predictive models utilizing machine learning algorithms such as Random Forests and Gradient Boosting Machines in R, Python and SQL, resulting in reduced downtime and optimized maintenance scheduling for industrial equipment.
- Integrated Generative AI Models into MLOps Pipelines: Designed and deployed Generative AI models, such as Large Language Models (LLMs) and Diffusion Models, into production using MLflow, SageMaker, and Google Vertex AI, automating model training, versioning, and deployment in a scalable and reproducible manner.
- Monitor model performance post-deployment and ensure optimal functionality.
- Set up logging, monitoring, and alerts for ML systems to track performance issues or errors.
- Automate ETL processes to ensure models have access to the right data.
- Build and maintain data pipelines for pre-processing and transforming raw data into usable formats.
- Ensure ML models and data comply with security policies and regulations.
- Implement data governance and secure access to sensitive customer and business data.
- Ensure ML systems are scalable, reliable, and can handle high demand periods.
- Used containerization and orchestration to simplify deployment and scaling of models across environments.
- Stayed updated on the latest ML and MLOps technologies and integrate them into workflows.
- Recommend tools and frameworks to improve productivity, scalability, and system robustness.
- Created detailed documentation for MLOps workflows, infrastructure, and deployment procedures.
- Maintain version control and ensure traceability of experiments and changes.
- Generated performance reports and provide visibility into ML operations for stakeholders.
- Offer ongoing support for production ML systems, troubleshoot issues, and resolve operational challenges.
- Managed incident response for ML production issues and implement improvements through post-mortems.

Professional Experience:

Amazon – India- Hybrid Data Scientist/ML-MLOps Engineer |July 2022 – Aug 2023

- Led the development of machine learning models for predictive analytics, customer segmentation, and recommendation systems.
- Used Scikit-learn, Tensor Flow, and PyTorch to design and optimize models for various business use cases.
- Conducted feature engineering, model tuning, and hyper parameter optimization to maximize model performance.
- Analysed large datasets using Pandas and SQL to derive actionable insights, improving decision-making for stakeholders.

- Optimized performance of LLMs by implementing model parallelism, distributed training, and leveraging multi-GPU or multi-node setups, ensuring cost-effective resource usage.
- Conduct exploratory data analysis (EDA) to derive actionable insights from large-scale healthcare datasets (e.g., patient data, prescription records, and claims).
- Set up and maintain MLOps pipelines to ensure the smooth deployment and lifecycle management of machine learning models in production.
- Use tools like Kubeflow and MLflow for model tracking, versioning, and experiment management, ensuring reproducibility and performance consistency.
- Implement model monitoring and alerting systems to track model performance, retrain models, and handle model degradation in real-time.
- Conducted data pre-processing and feature engineering tasks using Python and Teradata, preparing datasets for machine learning modelling.
- Write complex SQL queries to extract and manipulate data from multiple databases, including MySQL, PostgreSQL, and NoSQL databases like MongoDB.
- Generate detailed reports and dashboards using Tableau, Power BI, or Looker to support decision-making for internal teams and business stakeholders.
- Use Pandas and other tools to manage large datasets, optimize queries, and identify patterns in patient and prescription data.
- Visualized data trends and model results using Matplotlib, Seaborn, and Tableau for clear communication with business teams.

<u>Professional Experience:</u>

Cognizant – India - Remote Data Analyst | Nov 2021 – June 2022

- Performed data wrangling and exploratory data analysis (EDA) on large datasets to identify trends, patterns, and insights.
- Implemented A/B testing methodologies, resulting in a 20% improvement in marketing campaign effectiveness.
- Created dashboards and reports using Power BI and Tableau to support decision-making processes for stakeholders.
- Analysed large datasets to identify trends and insights, leading to a 15% increase in operational efficiency.
- Conducted market research and competitive analysis to inform strategic planning and business development.
- Collaborated with cross-functional teams to gather requirements and design solutions that address key business challenges.
- Collaborated with data scientists and engineers to document Python code and workflows, enhancing team knowledge sharing and onboarding processes.
- Assisted in the development of data models and predictive analytics to forecast future business trends and outcomes.
- Conducted A/B testing to evaluate the effectiveness of different strategies, providing data-driven recommendations for optimization

- Integrated machine learning models into CI/CD pipelines using Python-based tools (e.g., Jenkins, GitLab CI), facilitating seamless deployment and version control.
- Wrote complex SQL queries to extract, transform, and analyse data from relational databases.
- Collaborated with data science teams to support model development by preparing and cleaning data.